Apriori Algorithm

Program:

```
!pip install apyori import pandas as pd from apyori import
apriori ta = pd.read_csv("Market_Basket_Optimisation.csv",
header=None) transactions = [] for i in range(0, len(data)):
    transaction = []      for j in range(0, 20):    #
dataset has 20 columns max          item =
str(data.values[i, j])          if item != 'nan':
            transaction.append(item)
transactions.append(transaction) rules =
apriori(transactions,
min_support=0.003,      # 0.3%
min_confidence=0.2,     # 20%
min_lift=3,                  min_length=2)
results = list(rules)  for rule in results:
    items = list(rule.items)      # Extract rule parts
base = items[0]      support = rule.support      for
ordered_stat in rule.ordered_statistics:
antecedent = list(ordered_stat.items_base)
consequent = list(ordered_stat.items_add)
confidence = ordered_stat.confidence         lift =
ordered_stat.lift          if len(antecedent) > 0 and
len(consequent) > 0:             print("Rule:",
antecedent, "→", consequent)
print("Support:", round(support, 4))
print("Confidence:", round(confidence, 4))
print("Lift:", round(lift, 4))             print("-" *
40) from mlxtend.preprocessing import
TransactionEncoder te = TransactionEncoder() te_data =
te.fit(transactions).transform(transactions) df =
pd.DataFrame(te_data, columns=te.columns_)  #
Correlation matrix correlation = df.corr()  print("\n--
--- CORRELATION ANALYSIS -----\n")  # Flatten
correlation values corr_pairs = correlation.unstack()
# Remove self-correlation corr_pairs =
corr_pairs[corr_pairs < 1]  # Sort correlations
sorted_corr = corr_pairs.sort_values(ascending=False)
print("Top Positive Correlations:\n")
```

```
print(sorted_corr.head(10))  print("\nTop Negative
Correlations:\n") print(sorted_corr.tail(10))
```

Output:

```
Rule: ['light cream'] → ['chicken']
Support: 0.0045
Confidence: 0.2906
Lift: 4.844
----------------------------------------
Rule: ['mushroom cream sauce'] → ['escalope']
Support: 0.0057
Confidence: 0.3007
Lift: 3.7908
----------------------------------------
Rule: ['pasta'] → ['escalope']
Support: 0.0059
Confidence: 0.3729
Lift: 4.7008
----------------------------------------
Rule: ['fromage blanc'] → ['honey']
Support: 0.0033
Confidence: 0.2451
Lift: 5.1643
----------------------------------------
Rule: ['herb & pepper'] → ['ground beef']
Support: 0.016
Confidence: 0.3235
Lift: 3.292
----------------------------------------
Rule: ['tomato sauce'] → ['ground beef']
Support: 0.0053
Confidence: 0.3774
Lift: 3.8407
----------------------------------------
Rule: ['light cream'] → ['olive oil']
Support: 0.0032
Confidence: 0.2051
Lift: 3.1147
----------------------------------------
Rule: ['whole wheat pasta'] → ['olive oil']
Support: 0.008
Confidence: 0.2715
Lift: 4.1224
----------------------------------------
Rule: ['pasta'] → ['shrimp']
Support: 0.0051
```

```
Confidence: 0.322
Lift: 4.5067
----------------------------------------
Rule: ['avocado', 'spaghetti'] → ['milk']
Support: 0.0033
Confidence: 0.4167
Lift: 3.2154
----------------------------------------
Rule: ['milk', 'cake'] → ['burgers']
Support: 0.0037
Confidence: 0.28
Lift: 3.2114
----------------------------------------
Rule: ['turkey', 'chocolate'] → ['burgers']
Support: 0.0031
Confidence: 0.2706
Lift: 3.1035
----------------------------------------
Rule: ['milk', 'turkey'] → ['burgers']
Support: 0.0032
Confidence: 0.2824
Lift: 3.2384
----------------------------------------
Rule: ['frozen vegetables', 'cake'] → ['tomatoes']
Support: 0.0031
Confidence: 0.2987
Lift: 4.3676
----------------------------------------
Rule: ['tomatoes', 'cake'] → ['frozen vegetables']
Support: 0.0031
Confidence: 0.3651
Lift: 3.83
----------------------------------------
Rule: ['ground beef', 'cereals'] → ['spaghetti']
Support: 0.0031
Confidence: 0.6765
Lift: 3.8853
----------------------------------------
Rule: ['cereals', 'spaghetti'] → ['ground beef']
Support: 0.0031
Confidence: 0.46
Lift: 4.6818
----------------------------------------
Rule: ['ground beef', 'chicken'] → ['milk']
Support: 0.0039
Confidence: 0.4085
```

Lift: 3.152

Rule: ['milk', 'chicken'] → ['olive oil']
Support: 0.0036
Confidence: 0.2432
Lift: 3.6935

Rule: ['chicken', 'olive oil'] → ['milk']
Support: 0.0036
Confidence: 0.5
Lift: 3.8585

Rule: ['milk', 'olive oil'] → ['chicken']
Support: 0.0036
Confidence: 0.2109
Lift: 3.5161

Rule: ['spaghetti', 'chicken'] → ['olive oil']
Support: 0.0035
Confidence: 0.2016
Lift: 3.0604 --
Rule: ['chocolate', 'frozen vegetables'] → ['shrimp']
Support: 0.0053
Confidence: 0.2326
Lift: 3.2545

Rule: ['shrimp', 'chocolate'] → ['frozen vegetables']
Support: 0.0053
Confidence: 0.2963
Lift: 3.1084

Rule: ['chocolate', 'herb & pepper'] → ['ground beef']
Support: 0.004
Confidence: 0.4412
Lift: 4.4902

Rule: ['soup', 'chocolate'] → ['milk']
Support: 0.004
Confidence: 0.3947
Lift: 3.0462

Rule: ['cooking oil', 'ground beef'] → ['spaghetti']
Support: 0.0048
Confidence: 0.5714
Lift: 3.282

```
Rule: ['cooking oil', 'spaghetti'] → ['ground beef']
Support: 0.0048
Confidence: 0.3025
Lift: 3.079
----------------------------------------
Rule: ['ground beef', 'eggs'] → ['herb & pepper']
Support: 0.0041
Confidence: 0.2067
Lift: 4.1785
----------------------------------------
Rule: ['herb & pepper', 'eggs'] → ['ground beef']
Support: 0.0041
Confidence: 0.3298
Lift: 3.3565
----------------------------------------
Rule: ['eggs', 'red wine'] → ['spaghetti']
Support: 0.0037
Confidence: 0.5283
Lift: 3.0343
----------------------------------------
Rule: ['ground beef', 'french fries'] → ['herb & pepper']
Support: 0.0032
Confidence: 0.2308
Lift: 4.6658
----------------------------------------
Rule: ['herb & pepper', 'french fries'] → ['ground beef']
Support: 0.0032
Confidence: 0.4615
Lift: 4.6974
----------------------------------------
Rule: ['green tea', 'frozen vegetables'] → ['tomatoes']
Support: 0.0033
Confidence: 0.2315
Lift: 3.3847
----------------------------------------
Rule: ['frozen vegetables', 'spaghetti'] → ['ground beef']
Support: 0.0087
Confidence: 0.311
Lift: 3.1653
----------------------------------------
Rule: ['milk', 'frozen vegetables'] → ['olive oil']
Support: 0.0048
Confidence: 0.2034
Lift: 3.0883
----------------------------------------
Rule: ['frozen vegetables', 'olive oil'] → ['milk']
```

```
Support: 0.0048
Confidence: 0.4235
Lift: 3.2684
----------------------------------------
Rule: ['soup', 'frozen vegetables'] → ['milk']
Support: 0.004
Confidence: 0.5
Lift: 3.8585
----------------------------------------
Rule: ['milk', 'tomatoes'] → ['frozen vegetables']
Support: 0.0041
Confidence: 0.2952
Lift: 3.0973
----------------------------------------
Rule: ['mineral water', 'shrimp'] → ['frozen vegetables']
Support: 0.0072
Confidence: 0.3051
Lift: 3.2006
----------------------------------------
Rule: ['frozen vegetables', 'spaghetti'] → ['olive oil']
Support: 0.0057
Confidence: 0.2057
----- CORRELATION ANALYSIS -----
 Top Positive Correlations:  ground beef
spaghetti               0.195688 spaghetti
ground beef             0.195688 herb & pepper
ground beef             0.172579 ground beef
herb & pepper           0.172579 olive oil
whole wheat pasta       0.144450 whole wheat pasta
olive oil               0.144450 pasta
mushroom cream sauce    0.139067 mushroom cream sauce
pasta                   0.139067 mineral water
ground beef             0.138041 ground beef
mineral water           0.138041 dtype: float64  Top
Negative Correlations:  butter            mineral
water       -0.056505 mineral water      butter
-0.056505 spaghetti          cookies            -
0.058171 cookies            spaghetti          -
0.058171                    frozen vegetables   -
0.064247 frozen vegetables  cookies            -
0.064247 cookies            milk               -
0.065900 milk               cookies            -
0.065900 mineral water      cookies            -
0.100963 cookies            mineral water      -
0.100963 dtype: float64
```